

Periodic Progress Report

Hikaya – AI-Powered Platform for Preserving Palestinian Memory and Heritage

Team Name: Beta

Report Title: Sprint 1 Completion Report

Report Date: 16/07/2026

Reporting Period: 10/07/2026 – 14/07/2026

Project Manager: Aws Hammad

Team Members:

Aws Hammad – Birzeit University

Doaa Hatu – Birzeit University

Noureddin Etkaidek – Birzeit University

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Sprint 1 Status: Completed and Uploaded to GitHub

1. Project Summary

Project Title: Hikaya – AI-Powered Platform for Preserving Palestinian Memory and Heritage

Short Description:

Hikaya is an AI-powered interactive storytelling platform designed to preserve, document, and revive Palestinian history, culture, oral history, and collective memory. The platform allows users to explore historical and cultural locations across Palestine through an interactive map. When a user selects a location, the system presents historical information, simplified AI-generated narratives, multimedia content, and user-contributed stories.

The main goal of Hikaya is to transform static historical and cultural information into an engaging digital experience that supports cultural awareness, tourism, education, and the preservation of Palestinian identity.

2. Sprint 1 Objective

Sprint 1 was created to move the project from the planning stage into the first technical implementation stage. The goal was not to complete the full platform, but to prepare the technical foundation that allows the team to continue development in parallel.

The main Sprint 1 objective was to prepare the first working foundation of Hikaya by setting up the backend, frontend, database, AI workflow, content files, and GitHub collaboration structure.

3. Progress During This Reporting Period

During this reporting period, the team completed the main Sprint 1 tasks and pushed the work to GitHub. The focus was on building the project foundation, preparing the repository structure, assigning clear responsibilities, and making sure that each sub-team delivered an initial output that can be used in the next implementation phase.

The work completed during this period includes:

- **GitHub Repository Organization:** The project repository was organized into clear folders for backend, frontend, database, AI, content, and documentation work. This makes the project easier to manage and allows every sub-team to work in its correct area.
- **Backend and Database Foundation:** The backend and database team prepared the initial structure required for the backend side of the project. This included the database planning and the initial backend architecture needed for the MVP.
- **Frontend Foundation:** The frontend work started by preparing the main frontend direction and the required pages for the MVP, including the map view, location details view, story mode, and user contribution form.
- **AI Smart Features Foundation:** The AI team prepared the initial direction for the narrative generation workflow, including how historical content will be processed, simplified, and transformed into user-friendly stories without inventing unsupported facts.
- **Content Preparation:** The content documentation work was completed for the three selected MVP locations: Old City of Jerusalem, Bethlehem, and Sebastia. The content was prepared to support AI testing, backend data modeling, and frontend display.
- **Task Tracking and Sprint Management:** Tasks were clearly distributed across the team, and progress was tracked through GitHub commits and repository updates.

4. Completed Sprint 1 Deliverables

The following deliverables were completed and uploaded to GitHub during Sprint 1.

Backend & Database Team

Assigned Members: Nouredin Etkaidek and Doaa Hatu

- **Database Schema Design:**

- Prepared the initial PostgreSQL database design for the Hikaya platform.
- Included the main tables required for the MVP:
 - * Locations
 - * Stories
 - * Users
 - * Media metadata
 - * Content Categories
- Defined the main relationships needed between locations, stories, media files, categories, and users.
- Uploaded the database design work to the `database/` folder on GitHub.

- **Backend Architecture and API Skeleton:**

- Prepared the initial backend project structure.
- Defined the core backend direction for handling locations, AI-generated stories, and user contributions.
- Prepared the foundation for the main MVP endpoints:
 - * Fetching locations
 - * Retrieving stories related to locations
 - * Submitting user contributions
 - * Supporting future AI story generation requests
- Uploaded the backend structure to the `backend/` folder on GitHub.

Frontend Team

Assigned Member: Amra Sawalha

- **UI/UX Wireframing and Page Planning:**

- Prepared the initial frontend direction for the MVP.

- Defined the main user-facing screens:
 - * Interactive Map View
 - * Location Details Page or Sidebar
 - * Story Narrative Mode
 - * User Contribution Form
- Focused the design direction on simplicity, cultural identity, easy navigation, and clear storytelling.
- **Frontend Environment Initialization:**
 - Prepared the initial frontend project structure.
 - Organized the frontend folders for pages, components, assets, styles, and API/service communication.
 - Uploaded the frontend setup work to the `frontend/` folder on GitHub.

AI Smart Features Team

Assigned Members: Hussein Jaradat and Arwa Al-Hajahja

- **Narrative Generation Workflow:**
 - Prepared the first workflow for transforming historical content into simplified narratives.
 - Defined the general processing flow:
 - * Receive reviewed historical content as input
 - * Clean and organize the text
 - * Simplify the language
 - * Convert the content into an engaging story format
 - * Return the final narrative to the backend or frontend
 - Added rules to reduce hallucination risk by making the AI use only reviewed and provided content.
 - Uploaded the AI workflow documentation to the `ai/` folder on GitHub.
- **Backend Integration API Contract:**
 - Defined the initial JSON structure for communication between the backend and AI module.
 - The planned input structure supports:
 - * Location name

- * Historical text
- * Target audience
- * Language
- * Story tone
- The planned output structure supports:
 - * Final narrative
 - * Short summary
 - * Suggested title
 - * Content notes or uncertainty warnings if needed
- Uploaded the AI-backend contract to the `ai/` folder on GitHub.

Content Documentation Team

Assigned Member: Tamara Hamdan

- **Historical Data Gathering for MVP Locations:**

- Collected and organized historical and cultural information for the three selected MVP locations:
 - * Old City of Jerusalem
 - * Bethlehem
 - * Sebastia
- The prepared content includes:
 - * General overview
 - * Historical background
 - * Cultural and religious importance
 - * Important landmarks
 - * Markets, crafts, food, and traditions
 - * Popular stories and oral heritage when available
 - * Current heritage preservation challenges
 - * References and sources

- **Content Drafts Upload:**

- Formatted the collected information into organized content files.
- Uploaded the content drafts to the `content/` folder on GitHub.

- These files will be used as immediate testing data for the AI, backend, and frontend teams.

5. GitHub Repository Updates

The GitHub repository was used as the main place for tracking and storing Sprint 1 work. All completed Sprint 1 deliverables were pushed to the repository.

Repository Structure Updated During Sprint 1:

- `README.md`: General project description and overview
- `docs/`: Reports, sprint plan, task assignment, and documentation
- `frontend/`: Frontend project structure and UI planning
- `backend/`: Backend project structure and API planning
- `database/`: Database schema and database design files
- `ai/`: AI workflow and AI-backend contract documentation
- `content/`: MVP location content files

GitHub Workflow:

- The team used GitHub to track project progress.
- Work was organized into the correct project folders.
- Descriptive commits were used to show the progress of different sub-teams.
- The repository now contains a clear foundation for continuing implementation in Sprint 2.

Repository Link: <https://github.com/REPLACE-WITH-YOUR-REPOSITORY-LINK>

6. Sprint 1 Evaluation

Overall, Sprint 1 was successful because the team moved from planning into practical implementation and GitHub-based collaboration. The completed work provides a strong foundation for the next phase of the project.

What went well:

- The team had a clear sprint plan and strict deadline.
- Tasks were divided clearly between backend, frontend, AI, and content.
- The GitHub repository became organized and active.
- The content team prepared real data for the selected MVP locations.

- The AI direction became clearer and more connected to the system.
- The backend, frontend, and database work now have a starting structure for implementation.

What can be improved:

- The next sprint should focus more on integration between teams.
- The backend and frontend teams should agree on exact API response formats.
- The AI team should test its workflow using the prepared content files.
- The content files should later be refined into shorter AI-ready summaries.
- GitHub commits should continue regularly so the progress remains visible.

7. Challenges and Risk Management

- **Challenge: Time Pressure** Sprint 1 had a short deadline, so the team focused on foundation work rather than completing full features. **Mitigation:** The work was divided into smaller deliverables that could be completed and uploaded to GitHub.
- **Challenge: Integration Between Sub-Teams** Backend, frontend, AI, and content work are connected, so lack of coordination could slow the project. **Mitigation:** Sprint 2 will focus on connecting these parts through clear API contracts and shared data formats.
- **Challenge: AI Accuracy** AI-generated historical content may become inaccurate if it is allowed to generate unsupported facts. **Mitigation:** The AI workflow is designed to use reviewed historical content only and avoid inventing information.
- **Challenge: Content Size** Some historical content, especially for the Old City of Jerusalem, is very long and detailed. **Mitigation:** The full content will remain available, while shorter AI-ready summaries will be prepared later for testing and generation.

8. Plan for Sprint 2

Sprint 2 will focus on moving from project foundation into an integrated MVP prototype.

Planned Sprint 2 Tasks:

- **Backend & Database:**
 - Connect the backend with the database.

- Add seed data for the three MVP locations.
- Implement working APIs for locations, stories, and contributions.
- Prepare the backend endpoint for AI story generation.
- **Frontend:**
 - Build the first usable version of the home page and map page.
 - Display the three MVP locations.
 - Create the location details page.
 - Connect frontend pages with backend APIs.
- **AI Smart Features:**
 - Test the narrative generation workflow using the prepared content files.
 - Generate sample stories for at least one MVP location.
 - Support different story styles such as student, tourist, and child-friendly modes.
 - Prepare the first working AI function or service.
- **Content Documentation:**
 - Refine the existing content files.
 - Prepare shorter AI-ready summaries for each MVP location.
 - Add clearer source links where possible.
 - Add multimedia links or image references if available.
- **Project Management:**
 - Continue monitoring GitHub activity.
 - Create Sprint 2 task assignments.
 - Review pull requests and commits.
 - Coordinate integration between backend, frontend, AI, and content.

9. Conclusion

During Sprint 1, Team Beta successfully completed the first technical and organizational foundation for Hikaya. The team prepared the GitHub repository, uploaded the required project structure, completed initial backend and database planning, started frontend

organization, prepared the AI workflow and backend integration contract, and uploaded content drafts for the three selected MVP locations.

This sprint was an important step because it transformed the project from an idea and proposal into a structured technical project with visible progress on GitHub. The next step is to build on this foundation and begin integrating the backend, frontend, AI, and content components into the first working MVP prototype.

Prepared by: Aws Hammad

Role: Project Manager, Team Beta

Project: Hikaya – AI-Powered Platform for Preserving Palestinian Memory and Heritage